Nullius: auditing a claim of autonomous scientific discovery

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Abstract

Large language model (LLM) agents are increasingly reported to have *discovered* scientific results rather than merely to have executed them. Such attributions are hard to evaluate, because the artefact produced by discovery and the artefact produced by retrieval-plus-execution are the same artefact. We set out five minimum reporting and control requirements that would make the distinction assessable, and apply them to a recent, prominent claim: an LLM agent reported to have autonomously proposed and experimentally validated a previously unreported optical mechanism, an "optical bilinear interaction" analogous to the compatibility operation in Transformer attention (arXiv:2604.27092). Working entirely from the authors' released data and analysis code, we reproduce the released benchmark accuracies to within 5×10^{-4}. We then show that both experiments carrying the computational conclusion are scored under sample-level splits that place repeated measurements of the same physical pair in training and test. Re-evaluating with the pair as the unit of generalisation, and adding an exact permutation null over all 105 admissible category assignments ($p = 102/105$), a self-pair ablation, and a mirror-pair grouping, we find that the released evaluations do not distinguish a relation-specific signal from pair identity together with structural cues in the label design. The benchmarks remain valid demonstrations that repeated measurements from a fixed, previously observed pair library are decodable; they do not, under the released evaluation protocol, establish relation-specific structure, semantic organisation, or transfer to unseen pairs. Separately, the public record contains no backbone-model disclosure, agent trace, intervention log or architectural ablation, so the attribution of the scientific interpretation to an autonomous agent is not assessable from it—independently of the benchmark findings. We find no evidence in the inspected data that measurements were fabricated or internally inconsistent; the reported hardware automation may be substantial and is not evaluated here, and the two preceding studies in the target work were not audited.

Keywords: benchmark integrity; unit of generalisation; data leakage; permutation testing; shortcut learning; reproducibility; autonomous scientific discovery; optical computing.

1 Introduction

LLM-based agents have moved, over a short period, from assisting predefined scientific workflows to being credited with scientific results in their own right. The claims made on their behalf have escalated accordingly: not that a system executed a protocol competently, but that it *discovered* something that was not previously known.

This escalation has outrun the evidence customarily supplied to support it, for a structural reason. *The artefact produced by genuine discovery and the artefact produced by retrieval-plus-execution are the same artefact.* An agent that reasons from physical first principles to a new observable; an agent that recombines two textbook techniques it has encountered thousands of times in pretraining; an agent that retrieves a published protocol; and an agent steered by its operators toward a direction they found promising—all four emit the same thing at the end. Working code, a measured effect, a coherent manuscript. Nothing in the output separates them.

For human authorship this problem is bounded, and the bounding is so thoroughly built into scientific convention that it is easy to forget it is there. A human author's prior knowledge is finite, largely declarable, and traceable through citation; the reader can ask what was known and when. For an LLM agent the prior is a training distribution that is undisclosed, effectively unbounded, and—in precisely the cases where discovery claims are most attractive—saturated with the material at issue. Attribution therefore cannot rest on the output. It has to rest on controls and disclosures that were never previously required, because they were never previously necessary.

We propose five such requirements (Sec. 1.1) as a minimal starting set of reporting and control obligations, and then apply them to a recent and prominent claim (Secs. 2–5). The application is instructive beyond the individual case: two of the failures we find are invisible to ordinary peer review, because they live in the split logic and the label construction of a benchmark rather than in any statement the manuscript makes. They are, however, immediately visible to anyone who runs the released code with one line changed—which is the argument for requiring that the code be released, and for looking at it.

We state at the outset what this exercise is not. We allege no misconduct, we find no evidence of fabrication in the data we inspected, and we take no position on the reported hardware automation, which we did not audit. Section 1.4 makes the boundaries explicit.

1.1 Audit criteria

We audit against five requirements, referred to below as PROBE and stated in full in Appendix B: *Provenance* disclosure sufficient to make a causal attribution formulable; *Replication* across independent runs; *Operator-matched* controls, in which the comparison is the same quantity under the same operator rather than a different quantity; *Benchmark* integrity, meaning that the evaluation possesses the structure it purports to measure and that its partitions respect the unit of generalisation; and *Evaluation* pre-registration, or full disclosure of the search when choices were not fixed in advance. They are proposed as a minimal starting set, not a sufficient one.

1.2 Summary of findings

Table 1 states each claim we examined, what the public material establishes, the test applied, and—in the final column—the conclusion that survives our audit. That column is not a courtesy. Several of the original observations remain intact under our re-analysis, and the distinction between what survives and what does not is the substance of this paper.

1.3 Scope and unaudited claims

This audit is deliberately claim-specific. We examine the released Result-4 pipeline because it provides the empirical basis for the manuscript's claims concerning optical bilinear computation, semantic pair representation, and the analogy to a core operation in Transformer attention. We reproduce and stress-test the public data, scripts, feature construction and evaluation protocol associated with that result.

We do not present a complete audit of the two preceding studies. In particular, we have not independently evaluated the optical acquisition efficiency or the focusing-enhancement deficit in the transmission-matrix reproduction (Result 2), nor have we reconstructed the full theory-to-measurement argument, mixed-state synthesis or uncertainty propagation in the majorization-order study (Result 3). We therefore draw no conclusion about the correctness, novelty or autonomy of those results. Their inclusion in the target manuscript does not, however, resolve the benchmark-integrity failures identified for Result 4, nor does it provide evidence for the specific computational claims tested here.

Table 1: Claims examined, tests applied, and what still stands.

Claim	Public material shows	Audit test	Outcome (finding)	Minimum conclusion that stands
XOR interaction	sample-level 1.000	pair, mirror and token grouping	not retained under any pair-respecting split	repeated measurements of the 16 fixed pairs are decodable
Semantic relation	high sample-level accuracy	grouped CV plus exact label permutation	designated partition not exceptional (102/105, exact)	pair identity is decodable
Task D transfer	Complex-B above chance	identity rule and 2×2 ablation	above-chance value present only with self-pairs on both sides	a decodable difference exists under the fixed protocol
Attention analogy	a bilinear observable	operator comparison	weak form only	a fixed physical pair-feature primitive
Physical novelty	a measurable cross term	scoped bibliographic comparison (Sec. 4.1)	demodulation ingredients have established precedent; priority of the specific combination unresolved by this audit	the cross term is measurable on the platform
Autonomous discovery	trajectory narrative and resource counts	provenance and ablation audit	not assessable from the public record	the automation may be substantial; this audit does not evaluate it
Results 2 and 3	transmission-matrix and majorization studies	—	**not audited**	no conclusion drawn either way

Our operator analysis is similarly bounded. We distinguish identities that follow from the ideal shared-operator model from properties of the released experimental estimator. The empirical deviations reported in Sec. A do not identify their physical cause. They may arise from unequal propagation operators in the two input arms, residual self-reference terms, imperfect phase stepping, calibration error, drift, or another unmodelled component. Resolving these alternatives requires measurements and hardware information not present in the public deposit.

Finally, statements about absent evidence refer only to the public record available to us as of 30 July 2026: arXiv:2604.27092v1 and the Zenodo deposit 10.5281/zenodo.19890402. The deposit states that the formal Supplementary Information submitted to the journal, the Qiushi Engine core system and the hardware-control software are not included. We therefore distinguish "not present in the public materials" from "not performed". In particular, we cannot determine whether an operator-matched reconstruction, additional agent-provenance records or alternative evaluation protocols exist in non-public submission materials.

1.4 What we do not claim

We find no evidence in the inspected Result-4 data that the hardware measurements were fabricated or internally inconsistent. The raw frames, repeat structure and drift monitoring are substantial and mutually consistent. We make no assessment of the reported hardware automation—sustained agent control of a real optical platform over hundreds of steps—because the materials required to evaluate it are not public and we did not audit it; our findings neither support nor challenge it. Our claim is narrower and, we think, more serious: the specific evidence offered for the specific central computational conclusion does not support it.

2 Reproduction and evaluation design

The target work couples an LLM multi-agent system to a free-space optical platform and reports three studies of increasing autonomy, culminating in an open-ended investigation that identifies an "optical bilinear interaction". Two token-dependent fields E_a, E_b are coherently superposed with controlled relative phase, scattered by a fixed medium T and detected by a square-law camera; four-step phase-shifting demodulation with blank subtraction isolates

$$B_m(a, b) \;=\; E_a^\dagger \left(T_m^\dagger T_m\right) E_b, \tag{1}$$

a per-channel complex bilinear response. The computational claim is supported by an eight-token "semantic benchmark" comprising 64 ordered pairs and four tasks: pair identity (A, 64-way), same category (B, binary), category pair (C, 16-way), and cross-split same-category generalisation to a held-out category (D).

All material below derives from the authors' own deposit (DOI 10.5281/zenodo.19890402). Our pipeline mirrors `05_result4_refine_paper/scripts/analyze_advantage_boundary.py` so that any difference is attributable to a single documented modification.

Pre-specification of our own analyses. Since we ask the original work for pre-registration (Sec. 5.2), we state our own status plainly. The XOR audit of Sec. 3.6 was pre-specified: the ten tests, the splitting regimes and the reporting rule that all three possible outcomes would be published were written into the repository README and committed before the analysis was run. The semantic-benchmark analyses of Secs. 3–A were *not* pre-specified in that sense; they were exploratory, developed while reading the released code. We mitigated this in three ways—exact replication before any modification, exact rather than sampled null distributions where the label space could be enumerated, and adversarial self-checks against the objections we could anticipate—but the distinction is real and we do not blur it. Readers should weight the pre-specified XOR result accordingly.

2.1 Replication

We first reproduce the released numbers without modification. For the raw-input baselines—features computed directly from $E(x), E(y)$ with no optical propagation—the maximum deviation between our values and the published ones is 5×10^{-4} across all five feature families and all four tasks. The optical families reproduce likewise, and we independently recover the authors' participation-ratio effective ranks of the Complex-B feature ensemble: 3.00 at bin = 25 and 9.62 at bin = 10, against their stated $\sim$3 and $\sim$9. Nothing that follows is a pipeline discrepancy.

3 Benchmark findings

3.1 B1: the benchmark carries no semantic structure

The eight tokens are constructed as

```
def make_token_input(tid):
    seed = 10000 + tid * 7919
    rng  = np.random.default_rng(seed)
    phases = rng.uniform(0, 2*np.pi, (6, 6))
    return np.exp(1j * phases).flatten()      # 36 complex
```

that is, eight independent unit-modulus random phase vectors. The words (*cat*, *dog*, *hammer*, ...) and the categories (*animals*, *tools*, ...) are labels attached to random vectors; no embedding enters the pipeline. Consequently the proposition "*cat* and *dog* belong to the same category" is encoded nowhere in the inputs.

This much is common ground. The authors' own Report 4 lists the token–field mapping among its limitations and states that the mapping is arbitrary, in that "cat" and "dog" share no more optical structure than "cat" and "hammer". We take that premise from them. Where we differ is in the inference drawn from it: the report concludes that the observed semantic-category separation therefore arises from the interaction physics rather than from a semantically informed encoding. That inference presupposes that a category separation is present to be explained. Sections 3.3 and 3.4 test precisely that presupposition.

The immediate consequence of the shared premise is narrower but firm: the data-generating process contains no designed semantic relation from which held-out-category generalisation could be learned, so any above-chance Task-D result must be attributed to a finite-sample accident or to an evaluation cue, rather than to encoded semantics recovered by the measurement.

3.2 B2: Task D is accounted for by self-pairs

For each held-out category the test set contains 4 distinct positive pairs, of which 2 are self-pairs (x, x), against 24 distinct negatives. The measurement-free identity-rule baseline "same category $:= (x = y)$" therefore attains balanced accuracy $\frac{1}{2}\left(\frac{2}{4} + \frac{24}{24}\right) = 0.750$. Every full-readout value displayed in Fig. 1 lies at or below this baseline. We do not extend the comparison either to values reported under the different fixed-holdout protocol of the main manuscript, or to the higher values the supplement obtains under searched sparse-tap and output-budget configurations; those are exactly the configurations whose selection timing the public record does not document (Sec. 5.2). Removing all self-paired samples leaves no evidence of above-chance transfer (Fig. 1; 0.4979 and 0.4948 at the two readout scales), identically at both, while the per-fold values are constant to four decimals in the unablated case—the signature of a deterministic structural cue rather than a learned relation. We report the ablated values as point estimates against a 0.500 reference; with 2 distinct positive pairs per fold after ablation we attach no confidence statement to their exact magnitude.

3.3 B3: Tasks A–C depend on repeat-level splitting

Leakage of this kind—correlated repeated measurements of the same unit distributed across folds—is a documented and widespread failure mode in machine-learning-based science [3], with quantified inflation reported when the grouping unit is ignored [4]. Throughout this subsection, "leakage" means leakage relative to a claim of out-of-pair or semantic generalisation. If the tasks were intended only to ask whether repeated measurements of an already-observed pair can be recognised, splitting repeats at random is legitimate; the tasks then simply cannot support claims about category structure or new pairs.

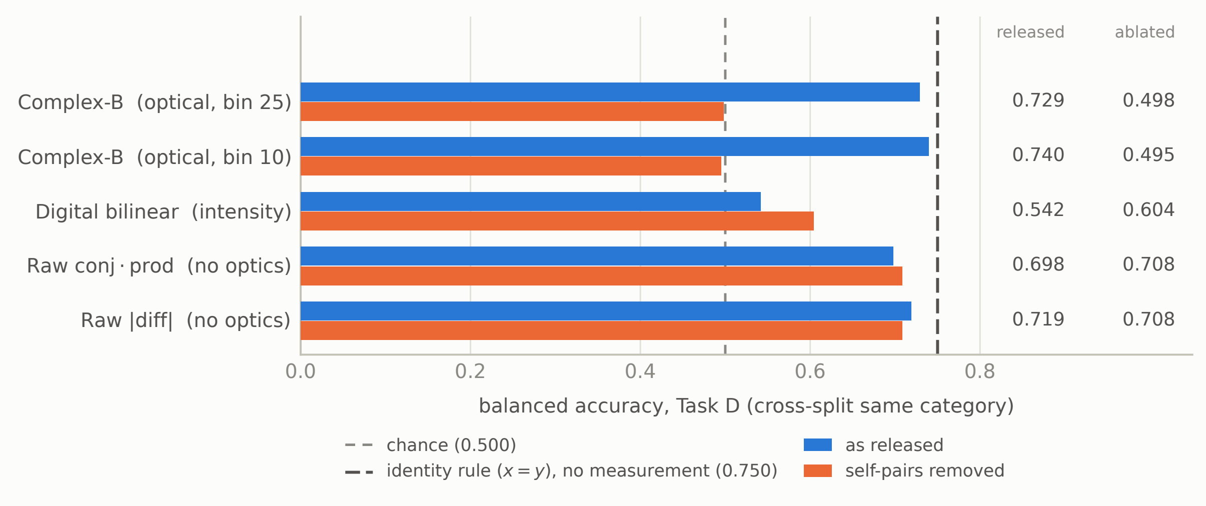

Figure 1: **Task D is accounted for by self-pairs.** Balanced accuracy on the cross-split same-category task under the refined held-out-category protocol, before and after removing every self-paired sample (x, x). The dashed rule at 0.750 is the accuracy of the measurement-free identity rule "same category := $(x = y)$", which uses the pair labels and no optical data at all. The optical Complex-B feature does not exceed that rule, and falls to chance once self-pairs are removed—identically at the coarse (bin = 25) and speckle-matched (bin = 10) readouts. The raw-input baselines, which involve no optical propagation whatsoever, are unaffected by the ablation.

The released scoring applies `StratifiedKFold` over 320 samples comprising 64 pairs × 5 physical repeats, so repeats of the same pair are split between training and test folds. We measure the pairwise repeat correlation of the Complex-B feature directly: mean 0.9828, with 97.0% of repeat pairs exceeding 0.97 (Fig. 3), confirming the authors' own figure. Training and test folds thus contain near-duplicates of the same measurement.

Re-scoring with `StratifiedGroupKFold` grouped by ordered pair gives Table 2 and Fig. 2. Pair identity is not evaluable as an out-of-pair generalisation task when the ordered pair is the grouping unit, because each 64-way class corresponds to exactly one group; and Task C falls to or below the 16-class chance level of 0.0625. We verified that the Task-C collapse is not an artefact of the fold count: across $n_{\text{splits}} \in \{2, 4, 5, 8\}$ every test class is represented in training (100%) and accuracy remains in 0.043–0.109 at both readouts (Fig. 2c).

Table 2: Tasks A–C at bin = 25, balanced accuracy, released scoring versus pair-grouped scoring.

Feature	Task	released	pair-grouped	Δ
Complex-B	A (64-way)	1.0000	degenerate by construction	
Complex-B	B (binary)	1.0000	0.6522	−0.3478
Complex-B	C (16-way)	1.0000	0.0474	−0.9526
Digital bilinear	B (binary)	1.0000	0.8917	−0.1083
Digital bilinear	C (16-way)	0.4437	0.1709	−0.2729

3.4 B4: the residual Task-B accuracy is not significant

Pair-grouped scoring leaves Task B at 0.6522, above the binary chance level. This residual admits an exact test, in the label-permutation family of [5]: because the admissible label assignments can be enumerated, the permutation null is exact rather than sampled. The same-category label is fixed entirely by a partition of the eight tokens into four unlabelled two-token categories, and

Figure 2: **Tasks B and C depend on repeat-level splitting.** Left, balanced accuracy at bin $= 25$ under the released scoring, which splits the five physical repeats of a pair across folds, and under pair-grouped scoring, which does not; the value column gives released $\to$ grouped for each feature family. Right, the Task-C outcome is not a fold-count artefact: it persists across $n_{\text{splits}} \in \{2, 4, 5, 8\}$ at both readouts, with every test class represented in training in all cases.

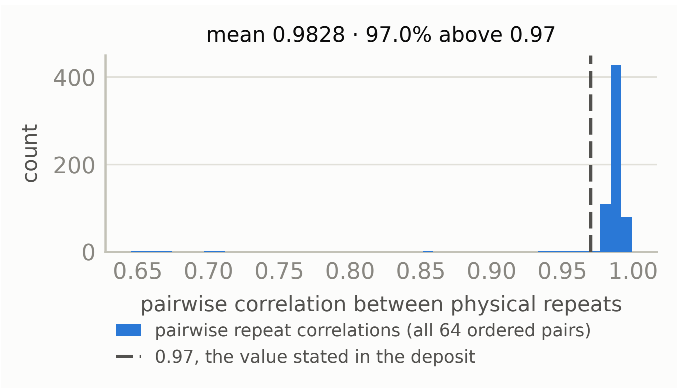

Figure 3: **Physical repeats are near-duplicates.** Distribution of pairwise correlations between the five repeated measurements of the Complex-B feature, over all 64 ordered pairs at bin $= 25$. Under the released scoring these near-duplicates are distributed across training and test folds.

there are $8!/((2!)^4 4!) = 105$ such partitions. We therefore recompute the pair-grouped Task-B accuracy under every admissible assignment. The designated semantic assignment is a member of this set, so the exact one-sided p-value is the fraction of assignments scoring at least as high as the designated one, with the designated assignment counted in the numerator; no Monte Carlo sampling and no continuity correction are involved.

The designated assignment scores 0.6522 against a null with mean 0.7280 and standard deviation 0.0347, and 102 of the 105 assignments score at least as high (Fig. 4); the exact one-sided p is 0.9714. Approximately 97% of all admissible category assignments therefore achieve accuracy at least as high as the designated semantic one. The residual is not evidence that the measurement recovers the designated category structure.

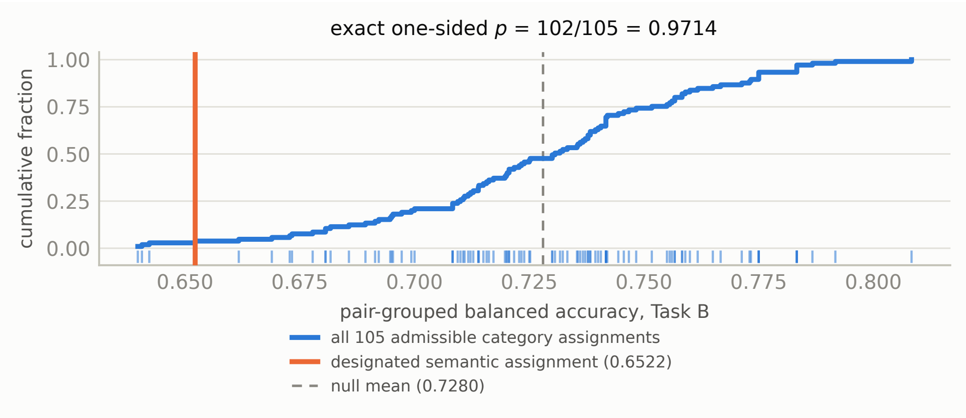

Figure 4: **The residual Task-B accuracy is not exceptional.** Empirical cumulative distribution and rug plot of pair-grouped Task-B balanced accuracy over *all* 105 partitions of the eight tokens into four unlabelled two-token categories—an exact enumeration, not a Monte Carlo sample. The designated semantic assignment lies in the lower tail.

3.5 B5: the Task-D shortcut is learned, not merely present

Removing self-pairs from the training set and from the test set are distinct interventions, and crossing them separates a cue that the classifier acquires during fitting from a structural shortcut sitting in the test set. Table 3 gives the full design. Above-chance performance occurs only when self-pairs are present on *both* sides; removing them from either side leaves no evidence of above-chance transfer. In particular, when self-pairs are present at test time but absent during training, accuracy is 0.4917—the shortcut is available but unused. This is a learned self-pair shortcut under this evaluation protocol, in the sense of [6]: a cue that is predictive within the evaluation as constructed, and that the classifier adopts in preference to the intended relation. We note the small effective sample: each held-out fold contributes only 4 distinct positive pairs (2 after ablation) against 24 distinct negatives.

Table 3: Task D at bin = 25 under the full 2×2 self-pair design. Identical classifier, folds and scoring throughout.

Self-pairs in training	Self-pairs in test	balanced accuracy	test +/−
present	present	0.7292	20 / 120
present	removed	0.4792	10 / 120
removed	present	0.4917	20 / 120
removed	removed	0.4979	10 / 120

3.6 B6: the XOR showcase tests repeat recognition, not relation generalisation

The four-token XOR showcase is the second experiment used to support the computational interpretation of Complex-B, and its logic differs from the semantic benchmark: an additive linear readout on concatenated single-token features genuinely cannot express a parity relation, whereas a product feature can. That identity is not in dispute. What we test is whether the released evaluation demonstrates a generalisable interaction feature.

The experiment contains all 16 ordered token pairs, each measured five times, with relation label $y_{\mathrm{XOR}}(x, y) = (x + y) \bmod 2$: eight positive and eight negative ordered pairs. The relation is symmetric, and all four self-pairs belong to the negative class. Parity here is a property of the token *index*; the tokens themselves are random phase patterns generated from unrelated seeds, so parity is not an encoded optical attribute.

We first reproduce the released sample-level evaluation exactly: Complex-B attains balanced accuracy 1.0000 on both 16-way pair identity and binary XOR classification. That split distributes the repeated measurements of every ordered pair across training and test folds. Because Complex-B also identifies the 16 observed pairs perfectly, the XOR score cannot distinguish learning the parity relation from recognising a pair and recalling its previously observed label. For a representation that resolves every pair, any fixed pair-to-label table is learnable; no XOR-specific geometry is required.

We therefore re-evaluate with the physical pair, rather than the camera repeat, as the unit of generalisation (Fig. 5). Because the number of distinct pairs is small enough that fold composition matters, we report both grouped K-fold estimates and exhaustive leave-one-group-out, pooling all out-of-fold predictions before scoring once—per-fold scoring is undefined here, since a held-out ordered pair contains a single class. Complex-B gives 0.5000 (ordered-pair K-fold), 0.6250 (unordered-pair K-fold), 0.5875 (exhaustive leave-one-ordered-pair-out), 0.3750 (exhaustive leave-one-unordered-group-out) and 0.4979 when a token is excluded from training entirely. We report these as point estimates against a 0.500 reference. Their test sets differ in size and in dependence structure, so we neither pool them nor attach a common uncertainty band, and their non-monotone ordering should not be interpreted. What they share is that the released value of 1.0000 is not retained once repeated observations of the test pair are excluded from training, and that no above-chance transfer is observed under any of them.

One grouped value does require explanation. The digital bilinear baseline reaches 0.8125 under ordered-pair grouping, above Complex-B. This is a leakage artefact with an exact mechanism: the feature $z(x) \odot z(y)$ is algebraically symmetric, so when (x, y) is held out its mirror (y, x) remains in training with a numerically identical feature vector. Removing the mirror as well collapses it to 0.0625. Complex-B, whose measured swap-conjugate correlation is 0.60 (Sec. A), leaks partially by the same route: 0.5875 under ordered grouping against 0.3750 once mirrors are grouped together.

Two caveats bound these numbers. First, with 16 distinct pairs and 80 samples, exhaustive leave-one-out is dominated by design artefacts: the concatenation baseline scores exactly 0.0000 under both exhaustive regimes, the signature of anti-learning induced by removing a sample from a balanced design rather than a meaningful below-chance result. Second, the task admits strong non-optical structural rules. Because every self-pair is negative, the measurement-free rule "negative if $x = y$, positive otherwise" attains balanced accuracy 0.7500; and if each token's parity group is read from its index, the relation is determined exactly with no optical measurement at all.

Finally, we enumerate all three ways to divide four tokens into two unlabelled groups of two. Under the same grouped evaluation the designated parity assignment scores 0.6250, equal to one alternative and below the remaining assignment at 0.7125. With only three admissible assignments this is descriptive rather than inferential, but it supplies no evidence that the designated parity structure is preferentially represented.

The mathematical statement that multiplicative features represent XOR where an additive linear readout cannot is not in dispute, and we do not claim that Complex-B lacks bilinear information. What the released experiment does not establish is that the measured field supplies a generalisable XOR interaction rather than a stable identifier of the 16 observed pairs.

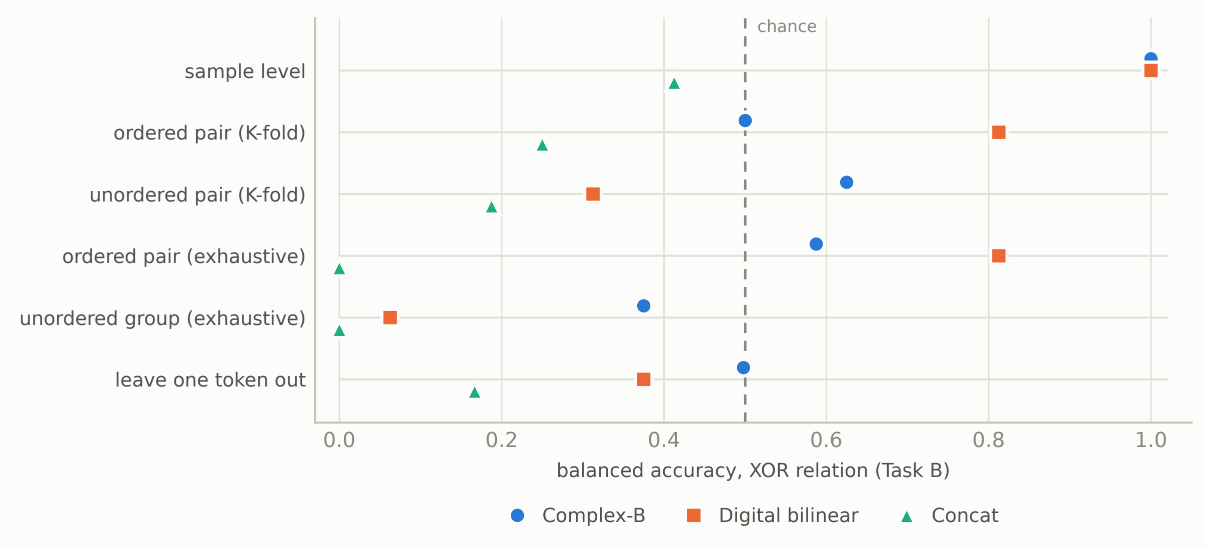

Figure 5: **The XOR result depends on the unit of generalisation.** Balanced accuracy on the parity relation under six splitting regimes, from the released sample-level split (repeats of a pair split across folds) to leave-one-token-out. The regimes are categorical and are deliberately not joined by lines; markers are offset vertically within each row so that coincident values remain visible—at sample level both bilinear families sit at 1.0000. The digital-bilinear value at exhaustive ordered-pair grouping is a mirror-pair leakage artefact: $z(x) \odot z(y)$ is symmetric, so the held-out pair's mirror carries an identical feature vector. Point estimates are shown against a 0.500 reference with no uncertainty band, since the regimes differ in test-set size and dependence structure.

3.7 The strongest reading available to the original evaluation

The findings above are not a claim that the reported accuracies are wrong. We reproduce every one of them. They are a claim about what those accuracies establish, and it is worth stating the authors' strongest position explicitly rather than leaving it to be raised in reply.

That position is that the released tasks were never intended to demonstrate generalisation to unseen pairs. On this reading, the sample-level split is an appropriate test of measurement stability and of linear decodability on a fixed, fully measured pair library; the XOR construction serves to illustrate the representational capacity of a bilinear feature relative to an additive linear readout; and the staged reports do qualify the claim, describing the result as holding "under the tested evaluation protocol".

We accept that reading, and under it the released numbers stand. Our point is what follows from it. A sample-level evaluation over a fixed pair library cannot separate three distinct hypotheses: that the feature encodes the relation, that it encodes pair identity from which a memorised label table is recovered, or that it exploits a structural cue such as the self-pair alignment. Because Task A shows that Complex-B resolves all 16—and all 64—observed pairs perfectly, the second hypothesis alone suffices to produce the reported relation accuracies, and no relation-specific structure in the measurement is required. The evidence is therefore consistent with the measurement being informative about the observed pairs, and does not discriminate among the mechanisms that could produce that.

This matters because the manuscript's interpretation does not stay at the level of observed-pair decodability. Claims about semantic relational structure, about pairwise relational computation, and about analogy to attention are all claims about a relation that transfers. That transfer is what the released evaluations do not test.

4 Computational interpretation

Equation (1) is the cross term of two-beam interference, isolated by four-step phase shifting. Under linear optics the jointly demodulated field and the digital product $A_x^* \odot A_y$ of the separately measured complex output fields A_x, A_y are the same quantity up to noise. The decisive control is therefore an equality test between the jointly demodulated field and that digital product, formed from the fields the *actual* two arms produce. We deliberately do not phrase this as a shared-T test: as Sec. A shows, the released estimator does not closely satisfy the shared-operator identities, so $A_x = T_x E_x$ and $A_y = T_y E_y$ with $T_x \neq T_y$ must be admitted. The shared-operator case is a special instance, not the general requirement.

The released controls are not this test. The "digital bilinear" baseline is $z(x)\,z(y)$ where z is binned camera *intensity*, discarding phase; a further variant is described in the supplement as a per-channel product "no cross-conjugation"; and the pseudo-B control replaces the physical T with a random complex Gaussian. The first two compare different algebraic objects; the third asks whether a real scattering matrix differs from a random one, not whether joint measurement yields information unavailable to separate measurement. The required control is absent, and—because this experiment records only single-token *intensity*—it cannot be performed on the released data. It can, however, be performed on the same apparatus: the self-referenced transmission-matrix measurement in the earlier studies of the same work recovers exactly the complex fields it requires.

4.1 Relation to established optical technique

This section is a scoped bibliographic comparison based on the principal methods and claims associated with these works; it is not an exhaustive, apparatus-level priority analysis. Each work below is cited for its headline contribution only, and we attach no finer device correspondence to any of them.

Two questions are worth separating, because the manuscript's abstract and its own staged reports answer them differently: whether the demodulation is new, and whether the use to which it is put is new.

On the first, the technique is established. Recovering an interference term from a small set of intensity frames taken at stepped reference phases is the basis of phase-shifting interferometry [7], and recovering a full complex field from intensity-only measurements by the same route is standard [8]. This is consistent with the authors' own characterisation of phase-stepping interferometry as a mature technique. The potentially distinctive element is therefore not the demodulation identity itself, but the particular combination of two encoded inputs, a fixed scattering transformation, channel-resolved recovery, and the interpretation of the result as a computational feature. Our search was not exhaustive and does not establish whether that combination is unprecedented.

On the second, each ingredient has substantial precedent in optical computing. Recovering complex per-channel quantities through a random scattering operator by phase stepping is how transmission matrices are measured [10]. Obtaining a product of two encoded optical fields by square-law detection is an established route to optical multiplication [11]. Using a fixed scattering medium as a random feature map is the basis of optical reservoir computing [9]. Establishing physical novelty therefore requires identifying which property of the combination was absent from prior optical correlators, scattering-based processors and coherent multiplication architectures—a determination the released materials do not make, and one that a systematic full-text prior-art review would be needed to settle. We record the position as *novelty not established*, which is distinct from novelty absent.

4.2 What is, and is not, implemented relative to attention

The manuscript describes the measured quantity as structurally analogous to a core operation in Transformer attention. We set out precisely how far that correspondence extends, because the claim is otherwise easy to read as stronger than the experiment supports.

An attention compatibility score is

$$s(x, y) \;=\; x^\top W_Q^\top W_K\, y, \tag{2}$$

while a measured detector channel supplies

$$B_k(x, y) \;=\; x^\dagger K_k\, y. \tag{3}$$

Both are two-input bilinear (respectively sesquilinear) compatibility forms, so the stated analogy holds in that weak mathematical sense, and we do not dispute it at that level.

The differences are the substance. W_Q and W_K are learned, may be distinct, and their product is a general matrix; K_k is fixed by the medium and is not adjustable at all. Under the ideal shared-operator model each unbinned K_p is a rank-one positive semidefinite outer product and a binned K_k is a finite sum of such terms, so the realisable operator family is a restricted subclass rather than a general learnable bilinear map—though, as Sec. A notes, if the two arms do not share an operator this particular restriction is weaker than it appears. Beyond the operator itself, the experiment does not construct a token-pair score matrix, and it implements no scaling, no softmax normalisation, no masking, no value projection and no value aggregation. No end-to-end attention task is run, and the readout is a linear classifier applied digitally to the measured features.

The defensible reading is therefore that the experiment demonstrates a restricted physical pair-feature primitive, not an implementation of Transformer attention and not attention hardware. Existing photonic-attention work supplies the relevant category distinction: implementing a bilinear primitive is not the same as implementing an attention layer, which ordinarily requires programmable projections, formation of a score structure, normalisation or an explicit substitute for it, and value-side processing. We do not enter those systems as comparanda here, since a like-for-like comparison would require throughput and energy figures that the target work does not report.

4.3 Scale, effective dimensionality, and unmeasured outlooks

Scale and effective dimensionality. The released computational demonstration contains eight fixed tokens and 64 ordered pairs, all derived from one deterministic random-phase codebook. It therefore does not test generalisation to a larger vocabulary, to newly encoded tokens, to another random codebook, or to an independently acquired physical operator. The target work itself identifies vocabulary scale as a limitation and notes, in the authors' released Report 4, that a related token-family representation previously exhibited degeneracy-boundary collapse at $N = 16$.

Nominal detector dimensionality should also be distinguished from effective dimensionality on the evaluated data. At the main-text bin size of 25 pixels, we reproduce a participation-ratio effective rank of 3.00 for the released Complex-B feature ensemble, despite hundreds of nominal detector channels; at bin size 10 the corresponding value is 9.62. These quantities are properties of the observed eight-token feature ensemble, not universal rank bounds on the apparatus. They nevertheless show that the high-dimensional capacity invoked in the computational interpretation was not demonstrated on the benchmark used to support it: most observed variation is concentrated in a small number of effective complex directions, and near-saturated accuracy is obtained on a correspondingly small, repeatedly observed vocabulary.

Speed and energy are unmeasured outlooks. The experiment establishes that the complex cross term can be recovered on the physical platform; it does not establish a system-level speed or energy advantage. In the released acquisition protocol, each ordered input pair is measured through four sequential phase settings. The overall protocol also contains blank-control measurements, camera readout, calibration, normalisation, demodulation, and digital classification. Blank controls may be reused across multiple pairs and therefore should not automatically be counted as a per-pair cost, but they remain part of the demonstrated system.

No end-to-end latency, pair-score throughput, energy per recovered score, total system power, or amortised calibration cost is reported in the public materials. Nor is the demonstrated implementation compared under a matched accuracy and precision target with a GPU, FPGA, or digital complex correlator. The current experiment acquires the ordered pairs individually; it does not demonstrate parallel construction of an $N \times N$ score matrix, softmax normalisation, value projection, or value aggregation. Future spatial, wavelength, or temporal parallelisation may change this assessment, but it is not measured here. Statements concerning high-speed or energy-efficient Transformer-like hardware should therefore be read as prospective motivation rather than experimental findings of the present study.

5 Discovery attribution and protocol status

The central claim is causal: that an agent, rather than its training distribution or its operators, originated the mechanism. Searching the full deposit we find no backbone model name or version, no system or initial prompts, no agent trace, no record of human intervention, and no direction-selection log; the deposit states explicitly that the engine and hardware-control software are not included. What is released—result-level data and offline analysis scripts—establishes that the numbers can be recomputed. It cannot establish where the idea came from.

We note a further tension internal to the released material. The staged reports characterise the contribution in measured terms: phase-stepping interferometry is “a mature technique”, and “the methodological novelty is not the demodulation itself but its application to extract a computationally meaningful bilinear observable”. The abstract of the manuscript describes the same object as a “previously unreported physical mechanism”. We record this as an internal inconsistency in how the contribution is characterised. It is not a completed novelty audit; Sec. 4.1 sets out how far the established literature reaches and what a determination would still require. Our position is that physical novelty is *not established* by the released materials, not that it is absent.

5.1 Architectural attribution

A separate attribution problem sits beneath the provenance one, and survives even if the trajectory were accepted as fully autonomous. The manuscript credits specific architectural components—Meta-Trace, the dual-layer separation, role–phase decoupling, accumulated research experience—with sustaining the long-horizon trajectory. A single successful run cannot identify which component, if any, produced the outcome. The released material contains no single-agent baseline, no run with Meta-Trace replaced by ordinary rolling summarisation, no fixed-workflow comparison, no ablation of the Critical Reviewer role, no run without inherited prior-study experience, no variation of random seed, and no variation of backbone model. Nor is there a human-expert time or cost baseline against which the reported speed advantage could be assessed. The reported totals—145.9 million tokens, 3,242 model calls, 1,242 tool invocations—are resource counts. They record what was spent, not which architectural choice was responsible for what was obtained.

5.2 Evaluation search space

The evaluation supporting the computational claim involves several free choices, and the released material shows that more than one value of most of them was examined. Table 4 records the search space that is publicly visible, alongside what the public record does and does not establish about when each choice was fixed. We stress the distinction: for none of these choices do the released materials document the selection *timing*, and we therefore do not assert that any of them was made after inspecting the outcome. What the record does establish is that the reported configuration was selected from a searched space without an independent confirmation set. This is adaptive evaluation, and the standard remedy—pre-registration of the readout parameters and splits, or a held-out physical realisation on which the selected configuration is evaluated once—was not applied.

Table 4: Publicly visible evaluation search space for the Result-4 benchmark.

Choice	Publicly observed search	Selection timing	Independent confirmation
Detector bin	5, 10, 25	Not publicly documented	None identified
Output budget	multiple n_{out} values	Not publicly documented	None identified
Task-D protocol	multiple descriptions and implementations	Protocol evolution observed	None identified
Feature family	multiple optical and digital families	Exploratory analysis evident	None identified
Token vocabulary	one fixed random codebook	Not publicly documented	No independent vocabulary

Protocol evolution and confirmatory status. The released reports document an evolving analysis rather than one fixed confirmatory protocol. Report 4 and Report 5 describe different cross-split constructions and report materially different behaviour across feature families. The later analysis additionally explores detector bin sizes, output-channel budgets, sparse-tap selections, raw-input controls, random-operator controls, and revised task definitions. Such exploration is scientifically useful and, when reported transparently, is not itself an error. Because the documented Task-D protocols are not identical, we do not treat numerical differences between the two reports as controlled method-to-method comparisons.

It does, however, change the evidential status of the resulting performance comparisons. The public materials do not document when the token vocabulary, task definitions, split rules, bin size, output budget, or highlighted operating point were fixed relative to inspection of the corresponding outcomes. No independently acquired confirmation set or new physical operator is used after these choices. We therefore interpret the released performance landscape as exploratory. A confirmatory computational claim would require a frozen protocol evaluated on new tokens, a new codebook, an independent acquisition, or another physical realisation without further selection of readout configuration.

6 Limitations and evidence needed

Before setting out what further evidence would help, we separate three things that this paper has deliberately kept apart, because reading them as one category would misstate our result.

Direct findings of this audit, each established by an ablation or an exact test on the released data: the mismatch between the evaluation unit and the claimed unit of generalisation; the fact that the designated category assignment is not exceptional among all 105 admissible assignments; the self-pair shortcut in Task D; and the mirror-pair reuse in the XOR showcase.

Absences in the public record, which are not findings about the underlying work: no operator-matched digital reconstruction, no backbone-model or prompt disclosure, no agent trace or intervention log, no architectural ablation, and no documentation of when the readout and split choices were fixed.

Outside our scope, on which we take no position: the transmission-matrix reproduction and the majorization-order study; the hardware performance of the platform; and an exhaustive priority determination for the measurement mechanism.

Our findings identify a failure of evidential attribution, not a failure to produce repeatable optical measurements. The released experiments show that the platform yields stable and decodable pair-dependent signals. They do not yet determine whether the decoded structure is relation-specific, whether the optical stage improves on an operator-matched digital reconstruction, or permit the origin of the scientific interpretation to be distinguished among model prior knowledge, agent-level synthesis and human steering.

The deficiencies are resolvable. They require different evidence for the computational, physical, and autonomy claims; no single additional accuracy value would address all three.

Agent provenance. Attribution of an idea to an autonomous agent requires disclosure of the backbone model and exact version, inference settings, initial and system prompts, retrieval sources, progressively disclosed skills, candidate directions, direction-selection decisions, and human interventions. The complete time-ordered Meta-Trace and tool trajectory should be released in a form that distinguishes contemporaneous records from summaries produced after the experiment. If safety, licensing, or instrument-access constraints prevent release of particular implementation details, the omitted fields and their relevance to the claimed discovery should be identified explicitly. Result-level data alone cannot establish the causal origin of the hypothesis.

Prior-availability control. The relevant question is not whether a current language model can reproduce the mechanism after publication, but whether the backbone available to the agent before the study already contained or could readily elicit the same proposal. This should be tested using the exact original model snapshot, without the agent scaffold or experimental feedback, from the same initial platform description. The prompt, decoding parameters, number of independent queries, retrieval permissions, and a prospectively defined scoring rule should be fixed before inspection of the outputs. The assessment should separately record whether the model proposes the interference cross term and whether it identifies the underlying phase-stepping and coherent-detection principles as established knowledge. Such a control would measure prior availability; it would not by itself prove the causal origin of the observed trajectory.

Operator-matched physical control. For each detector pixel, the ideal shared-operator model predicts

$$B_p(x,y) = (TE_x)_p^*(TE_y)_p. \tag{4}$$

For a binned channel $\mathcal{B}_k$, the corresponding reconstruction is

$$\bar{B}_k^{\text{digital}}(x,y) = \sum_{p\in\mathcal{B}_k} (TE_x)_p^*(TE_y)_p. \tag{5}$$

The decisive control is therefore to measure the two propagated complex fields separately under the same physical operator and compare this digital reconstruction, channel by channel, with the jointly demodulated $\bar{B}_k^{\text{physical}}$. Agreement should be quantified in amplitude, phase, residual structure, repeatability, and downstream performance under identical feature budgets and splits. If the two input arms implement different operators, the corresponding $T_x^\dagger T_y$ model should be stated and tested instead. This experiment would determine whether joint measurement supplies information beyond separately measured complex fields; intensity products and random-operator pseudo-B controls do not answer that question.

Benchmark with a valid unit of generalisation. A confirmatory benchmark should contain relational structure in the inputs rather than attach semantic labels to independent random vectors. It should use tokens, codebooks, or physical inputs not involved in configuration selection. All repeated measurements of the same physical condition must remain in the same fold whenever the claim concerns generalisation to new conditions. Self-pairs should either be excluded from relation evaluation or reported as a separate stratum, and performance should be compared with measurement-free structural rules and an exact or permutation null over admissible label assignments. Pair identity may be evaluated as repeat recognition, but it should not be interpreted as out-of-pair generalisation. The final test should use a new acquisition session, codebook, vocabulary, or physical operator after the evaluation protocol has been frozen.

Independent agent and hardware replications. A single successful trajectory cannot establish that the reported outcome is a reproducible property of the agent architecture. The open-ended study should be repeated independently from the same initial information, with all runs and unsuccessful directions retained. At least three complete runs would provide a minimal existence check, although more are required to estimate a meaningful discovery rate for a stochastic system. Replication should also separate agent stochasticity from hardware specificity by including, where feasible, an independent acquisition session or physical realisation. The reported outcomes should include the frequency with which the mechanism is proposed, selected, experimentally supported, rejected, or replaced by another direction.

Pre-registered confirmation. Before the confirmatory run, the token set, task definitions, train–test unit, treatment of self-pairs, detector binning, output budget, feature families, classifier, hyperparameters, primary endpoint, and statistical comparison should be frozen in a time-stamped protocol. Exploratory searches over bin size, n_{out}, task construction, or feature family may be reported in full, but the selected configuration should then be evaluated on untouched data without further adaptation. This separation would turn the released performance landscape from an exploratory result into a test of a specified computational claim.

Together, these additions would not require treating autonomous discovery as a binary label. They would allow readers to distinguish four progressively stronger conclusions: automated execution of a known protocol, recombination of available scientific knowledge, generation and testing of a new engineering interpretation, and experimental identification of a finding not available to the model or its operators beforehand.

A Model–measurement consistency

This section is diagnostic rather than evaluative. It bears on neither the benchmark failures of Sec. 3 nor the central attribution question, and we report it because it constrains what the released estimator can be assumed to satisfy.

Under the stated linear-optical model the unbinned channel kernel $K_p = T_p^\dagger T_p$ is *algebraically* rank-one, Hermitian and positive semidefinite; spatial binning forms $K_k = \sum_{p\in\mathcal{B}_k} K_p$, a positive semidefinite sum whose algebraic rank is bounded by the number of independent propagation rows in the bin. These are consequences of the model, not empirical assumptions, and they imply $B_p(y,x) = B_p(x,y)^*$.

What the release contains is an estimator carrying a reference arm, phase stepping, blank subtraction, drift and calibration error. It satisfies the ideal identities only approximately. Over the 28 off-diagonal token pairs the mean correlation between $\widehat{B}(y,x)$ and $\overline{\widehat{B}(x,y)}$ is 0.6042, against 0.1537 for the uncorrelated comparison $\widehat{B}(x, y)$; the conjugate symmetry is thus clearly present but far from exact. On self-pairs, where the model predicts a positive real value, the real part is non-negative in every channel, but the mean relative imaginary magnitude $|\mathrm{Im}\,\widehat{B}|/|\widehat{B}|$ is 0.3511.

A phase-gauge offset would reproduce this signature without implying that the cross term was not isolated, so we tested it: estimating a single global phase, and then a per-channel phase, from the self-pairs alone and applying it to all pairs. Neither accounts for the departure. The self-pair imaginary ratio falls only from 0.3511 to 0.3102 (global) and 0.3057 (per-channel), and the swap-conjugate correlation does not improve (0.5666 and 0.5606). The per-channel phases are moreover nearly identical (circular spread 0.0019), so there is no channel-dependent phase structure to remove.

We draw no conclusion about the cause. Two candidate explanations remain open and are not separable from the released data: a residual in the blank or reference subtraction, and the two input arms not sharing a transmission operator, $T_x \neq T_y$. We flag that the second would also weaken one line of criticism available against the attention analogy, since a kernel $T_x^\dagger T_y$ is rank-one but neither Hermitian nor positive semidefinite, and is in that respect *less* constrained than the shared-operator case. Resolving this requires acquisition-level information that the deposit does not contain.

B The Probe checklist in full

P — Provenance disclosure. The backbone model and version, its training cut-off relative to the publication dates of all target papers, the complete initial prompt, every candidate direction generated and discarded, the criterion by which a direction was selected and by whom, every human intervention, and the full agent trace. Without these, the causal attribution "the agent discovered X" is not merely unverified but unformulated.

R — Replication. At least three independent runs from the same starting prompt, with the distribution of outcomes reported. A single trajectory from a stochastic system cannot distinguish an architectural property from a sampling event.

O — Operator-matched control. Where a physical measurement is claimed to produce information, the control must be the digital reconstruction of the *same* quantity under the *same* operator, not a different quantity. Randomising the operator, or substituting an algebraically distinct expression, tests a different hypothesis.

B — Benchmark integrity. The evaluation must possess the structure it purports to measure; train/test partitions must respect the unit of generalisation; and every reported accuracy must be compared against the best measurement-free rule available on the same partition, and against an exact or permutation null over the admissible label assignments where one can be enumerated.

E — Evaluation pre-registration. Readout parameters, feature budgets and splits must be fixed before the comparison of interest, or the search over them must be reported in full.

Data and code availability

All analysis code, the audit modifications and the full result files are at `https://github.com/SII-WANGZJ/Nullius`. The audit consumes only the authors' public deposit (DOI 10.5281/zenodo.19890402); no data of our own were generated, and the deposit is not redistributed. Every file read from the deposit is recorded with its SHA-256 in `results/input_manifest.json`; the roll-up digest over all 1543 consumed files is `b273f7f6c8d5bd8d7bb849e1d9908ea35a8ec171b9870b7cd2cb27b496351350`, so any subsequent change to the deposit is detectable. The environment is pinned in `environment.yml` and `requirements-lock.txt`.

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